



Improved Fab presentation on phage surface with the use of molecular chaperone coplasmid system



Qiuting Loh, Siew Wen Leong, Gee Jun Tye, Yee Siew Choong, Theam Soon Lim *

Institute for Research in Molecular Medicine, University Sains Malaysia, 11800 Penang, Malaysia

ARTICLE INFO

Article history:

Received 31 October 2014

Received in revised form 24 February 2015

Accepted 24 February 2015

Available online 10 March 2015

Keywords:

Antibody phage display

Fab presentation

Coplasmid

Molecular chaperone

ABSTRACT

The low presentation efficiency of Fab (fragment antigen binding) fragments during phage display is largely due to the complexity of disulphide bond formation. This can result in the presentation of Fab fragments devoid of a light chain during phage display. Here we propose the use of a coplasmid system encoding several molecular chaperones (DsbA, DsbC, FkpA, and SurA) to improve Fab packaging. A comparison was done using the Fab fragment from IgG and IgD. We found that the use of the coplasmid during phage packaging was able to improve the presentation efficiency of the Fab fragment on phage surfaces. A modified version of panning using the coplasmid system was evaluated and was successful at enriching Fab binders. Therefore, the coplasmid system would be an attractive alternative for improved Fab presentation for phage display.

© 2015 Elsevier Inc. All rights reserved.

In phage display technology, antibodies presented on phage particles are often a smaller variant of the antibody molecule. The Fab format is less preferred compared to the smaller single chain variable fragment (scFv)¹ in phage display. The Fab format consists of a pair of heavy and light chains that is linked together via a disulphide bridge. During the presentation of antibody on the phage surface, the heavy chain will be presented as a fusion to the phage coat protein while the light chain is expressed individually [1,2]. The light chain is required to locate the heavy chain in the periplasmic region and form a disulphide bridge between both chains. This complex process can lead to incomplete Fab fragments being presented during phage packaging. This results in the inefficiency of complete Fab presentation on phage particles [1]. In order to improve the presentation efficiency of Fab on phage the improved efficiency of disulphide bond formation is vital.

In bacteria, DsbA and DsbC, which belongs to the “Dsb (disulphide bond)” family, are molecular chaperones that assist in disulphide bond formation. There are two possible pathways involved in Dsb protein folding and maturation: (i) an oxidation pathway and (ii) an isomerization pathway, catalyzed by the thiol-disulphide oxidoreductases DsbA and DsbC [3–5]. pTUM4 encodes for four protein-folding catalysts and carries two artificial dicistronic operons, one encoding FkpA and SurA and the other one encoding DsbA and DsbC [6]. The DsbA protein acts as a disulphide donor

while the DsbC functions as a disulphide isomerase [7–10]. The coexistence of these two proteins as the thiol-disulphide oxidoreductases allows the catalysis of the formation and isomerization of disulphide bridges. Besides the Dsb proteins, FkpA and SurA are also involved in the protein-folding pathway. These proteins exhibit as peptidyl-prolyl *cis/trans*-isomerases [11,12]. In addition, FkpA is also known as a global folding enhancer as it can assist in protein folding that does not involve PPIases activity. It suppresses the formation of insoluble proteins, improving the yield of protein expressed to the periplasm of *Escherichia coli*. SurA is involved in the assembly of the endogenous outer membrane porins and enables correct folding of the protein. It also prevents the formation of aggregated protein or truncated protein in the periplasm region. Therefore, both FkpA and SurA can increase the yield of the proper folded protein and improve cell viability [4,13,14]. With the concomitant overexpression of DsbA, DsbC, FkpA, and SurA, pTUM4 enhances the periplasmic secretion of functional recombinant proteins in *E. coli* [4]. Schlapschky et al. had reported that the coplasmid pTUM4, consisting of DsbA and DsbC, has shown improved formation of the disulphide bond during protein expression in *E. Coli*, resulting in better protein folding [3,4]. However, the role of these proteins during phage packaging is not well studied. Thus, the improvement of disulphide bridge formation in Fab fragment generation with the assistance of pTUM4 during phage packaging was investigated.

It is imperative that a Fab fragment presented on phage has a complete set of heavy and light chains. However, the Fab fragments are composed of interchain and intrachain disulphide bonds

* Corresponding author. Fax: +60 4 653 4803.

E-mail address: theamsoon@usm.my (T.S. Lim).

¹ Abbreviations used: Dsb, disulphide bond; scFv, single chain variable fragment.

[15]. Fab fragments derived from IgG and IgD are relatively different in the positioning of the interchain disulphide bridge. To illustrate this, the position of interchain disulphide bonds in IgG and IgD is shown in Fig. 1. The structural model for IgG was obtained from PDBID:1HZH while PDBID:1ZVO was the only available 3D structure for IgD to date. As 1ZVO only contained C α atoms, the full IgD structure was reconstructed with Modeller 9v10 using self-homology modeling [16,17]. Sequence alignment was performed with ClustalW2 [18]. The 1ZVO sequence was remodeled using its C α atoms positions as template. A total of 100 models were randomly generated and the model with the best discrete optimized potential energy (DOPE) score was chosen as the representative IgD model. Bond length was calculated between the sulfur atoms of disulphide bonding partners using the measure bond command in VMD [19].

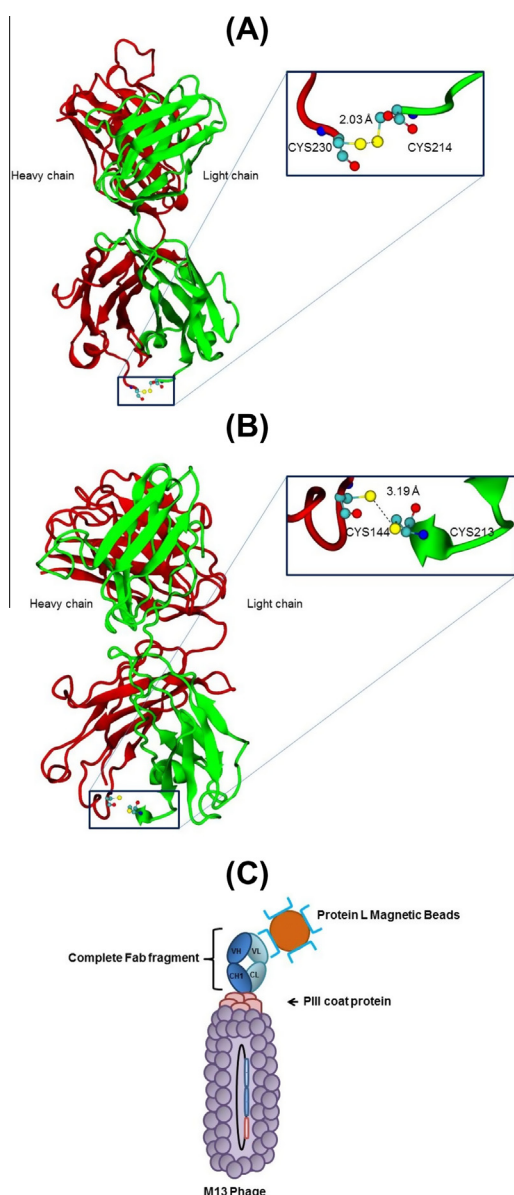


Fig. 1. The 3D structures of Fab format of an antibody. (A) Immunoglobulin G (IgG), (B) Immunoglobulin D (IgD). The heavy chain is depicted in red, while the light chain is in green. For IgG, the disulphide bond is formed by cysteines at residue 214 and 230, while for IgD, it is at residue 144 and residue 213. (C) Schematic of the proposed assay for the detection of the light chain of the Fab fragment using Protein L magnetic beads. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

We designed a new protocol to determine the successful presentation of a Fab fragment on phage. Detection of the kappa light chain was used as the indicator for a successful presentation of an intact Fab fragment. As the heavy chain is fused to the coat protein of the phage (pIII), only Fab fragments with an intact kappa light chain would be captured by the protein L magnetic beads, allowing easy separation of bound and unbound phage particles (Fig. 1).

Materials and methods

Construction of phagemids containing IgG and IgD

First, phagemids containing Fab fragments of IgG and IgD were constructed, respectively. The light chain of the Fab fragment used is the kappa light chain due to its specificity to protein L. The heavy chain constant region (CH1) of IgG and IgD was amplified using gene-specific primers from the total cDNA that was obtained from the collected blood sample under the approval of University Sains Malaysia Ethics Committee. The genes were subcloned to an in-house phagemid vector, pFAB1 (Supplementary Data). Sequencing of the clones was carried out to confirm the insertion of both the heavy and the light chains (1st Base, Malaysia). pTUM4 was then transformed into TG1 cells, cultured, and prepared as competent cells. The IgG and IgD phagemids were transformed into the TG1 cells harboring pTUM4 with antibiotic selection.

Phage packaging of IgG and IgD presenting phage

To ensure the presentation of complete Fab molecules, phage packaging of the IgG and IgD with pTUM4 was carried out as well as the packaging of IgG and IgD without pTUM4 as controls. The IgG and IgD clones with pTUM4 were cultured in 50 ml 2X YT broth with the addition of 0.1% glucose, 100 μ g/ml ampicillin, and 12.5 μ g/ml chloramphenicol while the clones without pTUM4 were cultured in the same amount of broth and antibiotic without chloramphenicol. These clones were cultured up to OD₆₀₀ = 0.5, and then infected with helper phage, M13K07, at 37 °C for 1 h. After the infection, 0.1 μ M IPTG (CalBioChem, USA) was added into the cultures, which were incubated overnight at 30 °C at 180 rpm in the incubator shaker (Thermo Scientific, USA). The phage particles produced were precipitated with PEG/NaCl and titration of the phage particles was carried out to ensure that the same amount of phage particles are used during the evaluation of Fab phage presentation.

Phage presenting Fab-capture assay using protein L beads

For each phage presenting Fab, 10 μ l of Protein L magnetic beads (Bioclone, USA) was used to capture the kappa light chains presented on the phage particles. The beads were blocked with 2% PTM (phosphate buffer saline with 1% Tween 20 and 2% skim milk) for 1 h, washed three times with 200 μ l of PBS-T (phosphate buffer saline with 1% Tween 20), and incubated in each phage clone solution for 1 h with shaking. The starting amount of phage particles at 1×10^9 cfu/ml was used for pull down. After the incubation, the beads were washed with 200 μ l of PBS-T for three times. The bound phage particles were then reinfect with 100 μ l of TG1 culture with an OD₆₀₀ = 0.5. The culture was incubated at room temperature for 30 min without shaking. A 1:10 dilution series of the culture was prepared in 100 μ l of 2X YT broth in order to determine the number of phage particles recovered. Ten microliters of each dilution was then spotted on 2X YT agar (2% glucose + 100 μ g/ml ampicillin) plates. These plates were then incubated overnight at 37 °C.

Further confirmation was done using ELISA (enzyme-linked immunosorbent assay) by detecting the amount of phage particles rescued using the magnetic beads. The same amount of phage presenting Fab molecules (1×10^9 cfu/ml) was coated on the microplate overnight and blocked with 2% PTM for an hour to avoid background. Then 100 μ l of 1:5000 dilution of Protein L-HRP (Thermo Scientific, USA) and anti-M13 HRP (GE Healthcare, USA) was added in each of the wells and incubated for an hour. All wash steps were carried out with 200 μ l of PBS-T thoroughly to remove unbound phage particles. Then 100 μ l of ABTS (Amresco, USA) was added and the absorbance was measured at 405 nm using a Multiskan spectrophotometer (Thermo Scientific, USA).

Phage display panning

Subsequently, we incorporated the use of the coplasmid in phage display panning against ubiquitin using a Fab antibody library with slight modifications (Fig. 2). First, antigen (ubiquitin) was coated on the microplate overnight and blocked with 2% PTM for 1 h. The preincubation of the in-house Fab phage library was carried out with 2% PTM for 1 h. The phage library was then transferred to the microplate and incubated for 1.5 h. Then, the unbound antibody phage was removed by washing with PBS-T. The bound Fab phages were eluted with 100 μ l of trypsin solution for 30 min at 37 °C. The eluted phages were added with the TG1 cells with the $OD_{600} = 0.5$ for 1 h at 37 °C and then the infected cells were allowed to culture for 3 h. After 3 h, plasmid extraction from the infected cells was carried out and then transformed into TG1 cells carrying pTUM4. These cells were plated on 2YT agar with 2% glucose, ampicillin, and chloramphenicol and incubated overnight at 37 °C. Next day, the cells carrying the phagemid and pTUM4 were scraped and inoculated into 100 ml of 2YT with

glucose, ampicillin, and chloramphenicol. The cells were grown for 2 h at 37 °C and 200 rpm. Then, the bacteria were infected with M13KO7 (1×10^9 cfu/ml) and incubated for 1 h at 37 °C without shaking, followed by 1 h at 37 °C with shaking (200 rpm). Then, the culture was centrifuged and the supernatant was removed. The pellet was then resuspended with 2YT broth with ampicillin, kanamycin, chloramphenicol, and IPTG. The culture was incubated overnight at 30 °C. Next day, the culture was centrifuged and the supernatant was precipitated with PEG/NaCl for 1 h. After precipitation, the phage was obtained and used for the next round of panning. To analyze the Fab phages obtained from the panning method, ELISA was carried out. The microplate was coated with antigen overnight and blocked for 1 h. As for control, the wells were coat with 2% PTM. Then, the phages were added into each well and incubated for 2 h. All the washing steps were carried out using PBS-T. Anti-M13 HRP (GE Healthcare, USA) was added into each well and incubated for 1 h. Then 100 μ l of ABTS (Amresco, USA) was added into each well and the absorbance measurement was taken using a spectrophotometer.

Results and discussion

During phage display, Fab constructs are expressed as two individual chains with either chain being fused to the pIII coat protein. In our system, the heavy chain is fused to the coat protein to ease our analysis process. The formation of a full Fab construct can only be achieved when both chains are transported to the periplasmic layer for disulphide bond formation [20]. Therefore, in order to transport both chains to the periplasmic region, two independent leader peptides are used to transport both chains to the periplasmic layer [21,22]. The disulphide bond formation between the two chains is not only for binding but also plays an important role

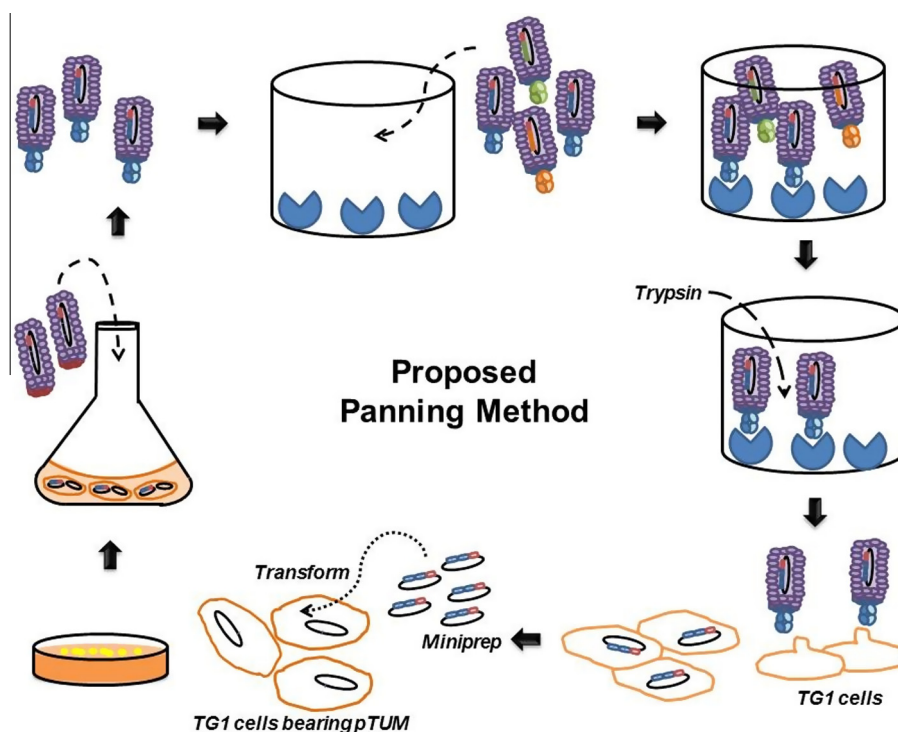


Fig. 2. Proposed phage display panning method. This method incorporates the coplasmid pTUM4 in the panning rounds. After the infection of the rescued phage, plasmid extraction of the infected cells was carried out and transformed into TG1 cells harboring pTUM4. The next day, the cells were scraped, inoculated into 2X YT broth containing glucose and antibiotics, and allowed to grow for 2 h. After that, helper phage was added into the culture. The infected cells were then centrifuged and resuspended with fresh 2X YT broth (Amp/Cam/Kan/IPTG). This process will be repeated for the subsequent rounds.

in terms of stabilization of the protein. The coplasmid pTUM4 has been reported to successfully improve the expression of soluble Fab fragments via improved formation of the disulphide bond in the bacterial periplasmic space. However, the system has never been used for improved phage presentation.

Taking into account the M13 phage replication cycle, the ability to apply the pTUM4 system for Fab phage packaging is similar to that of soluble Fab expression. When the pTUM4 coplasmid cells with the phagemid are infected with the helper phage, phage particles will initiate the phage replication cycle producing new phage particles with the Fab. The Fab gene fused to the pIII coat protein will assemble together with the other coat proteins in the inner membrane of the *E. coli* cells before exiting the cell without lysis [23]. The coplasmid proteins DsbA and DsbC that are also transported to the periplasmic region will be present to improve the disulphide bond formation in the periplasm during the final stage of phage assembly [4,9]. The light chain being expressed and transported independently to the periplasm will then have an improved opportunity to form the disulphide bridge with the pIII fused heavy chain. The pTUM plasmid also carries the gene for the peptidyl-prolyl *cis/trans* isomerase or FkpA and SurA. FkpA and SurA play a role in assisting the correct folding of protein in the periplasmic region [24]. The peptide-binding site of FkpA will transiently bind to the Fab fragment to assist in proper folding of the Fab. Therefore, FkpA and SurA together with the Dsb proteins could potentially work in tandem to improve the Fab presentation on phage particles.

As shown in Fig. 3A, the recovered phage presenting IgG and IgD on the magnetic beads with the coplasmid recorded approximately twofold higher titer compared to the phage particles without the coplasmid. For IgG and IgD with pTUM, the amount of phage packaged was estimated at 2×10^5 and 1.9×10^5 phage particles, respectively. On the other hand, for IgG and IgD without pTUM4, an estimated amount of 7×10^4 and 9×10^4 of phage was packaged, respectively. This showed that phage packaging under both conditions for both IgG and IgD was comparable. The introduction of the coplasmid to F' host cells did not interfere with the infection capabilities of the host cell as well as the phage packaging abilities of the cell. Therefore the host cell prepared with the pTUM4 was able to tolerate the presence of the coplasmid as well as the phagemid with the helper phage infection.

In order to examine the Fab presentation efficiency, an assay using Protein L was developed. Both IgG and IgD constructs were prepared with a kappa light chain. The affinity of Protein L toward the kappa light chain will allow the determination of the presence of the kappa light chain attached to the heavy chain. The assay was designed to detect only fully assembled Fab fragments on the beads. The signal generated would indicate that successful binding of both heavy and light chains was achieved. An increase in absorbance was recorded for both IgG and IgD Fab presenting phage when packaged with the coplasmid (Fig. 3B). A positive control using anti-M13 monoclonal antibodies (GE Healthcare, USA) was used to confirm successful packaging and coating of phage particles on the microtiter plates. The high absorbance readout from all samples using the anti-M13 antibody suggests that phage particles were well packaged and are bound on the solid phase. The inability of protein L-HRP to record a significant signal for packaged IgG and IgD phage fragments without the coplasmid indicates a lower presentation efficiency of an intact IgG and IgD Fab molecule. The absorbance reading for coplasmid IgG and IgD was significantly higher than the samples without the coplasmid. However, the absorbance values of IgG and IgD with the coplasmid are similar which indicated equal efficiency for both IgG and IgD.

IgG and IgD constructs share similar shapes but vary in terms of the disulphide bond linkage. The cysteines responsible for disulphide bond formation in IgG and IgD are located at different positions on the sequence. This resulted in a change in distance and

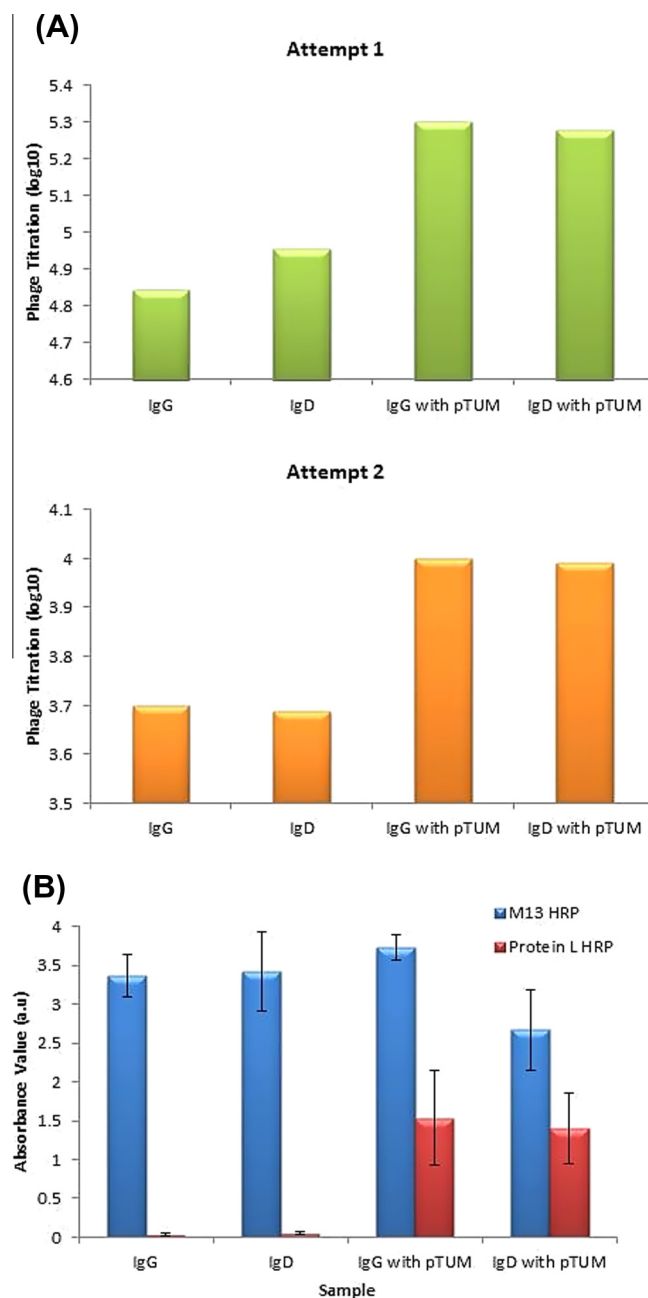


Fig. 3. Evaluation of the Fab phage presentation efficiency. (A) Titration of the recovered phage particles of IgG and IgD with and without pTUM after panning against Protein L magnetic beads. (B) Absorbance values of IgG and IgD presenting phage with and without pTUM.

length of the bonding. However, the similarity of presentation for both IgG and IgD Fab fragments using the coplasmid suggests that the positioning of the cysteine residues for disulphide bond formation and the distance between the heavy and the light chain are not limiting factors for disulphide bond formation. The chaperones are also capable of tolerating the difference in distance for the bond formation.

A comparison of the conventional plate panning process with the modified coplasmid panning protocol was also carried out. In order to investigate the influence of the coplasmid and chaperones in Fab formation, a panning experiment was carried out. We first conducted the panning procedure using the conventional method where rescued phages were infected with TG1 cells bearing pTUM4. The amount of rescued phage using this method was

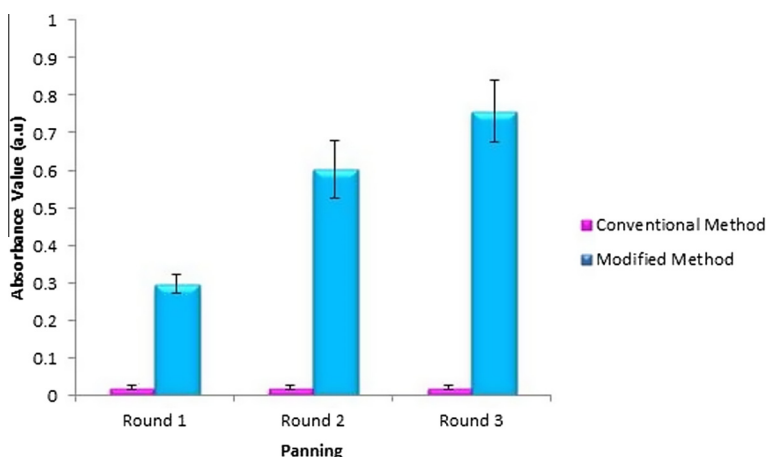


Fig. 4. Absorbance values of the polyclonal ELISA of Fab phage display panning rounds.

acceptable at 10^6 . However, we noted a 100-fold decrease of rescued phage using TG1-pTUM4 cells when compared with phage rescued using TG1 cells only. There seems to be a drop in packaging efficiency when direct infection of rescued phage to TG1-pTUM4 cells was carried out. However, we are unable to explain the reason for the decrease in rescue. The protocol for rescue was modified from the conventional panning method to allow better rescue using TG1 cells only for infection followed by culture for 3 h and recovery of phagemid DNA. The reason is to ensure no loss in antibody diversity as a result of the 100-fold lower packaging efficiency. The recovered phagemid was then transformed into TG1-pTUM4 cells for phage packaging. Introduction of the intermediary phagemid preparation step before packaging allowed for better phage packaging for the subsequent panning round.

The panning experiment with the conventional panning method that uses TG1-pTUM4 cells to rescue the bound phage showed no significant enrichment after three rounds of panning (Fig. 4). However, the proposed coplasmid-assisted panning method was able to show polyclonal phage enrichment after three rounds of panning (Fig. 4). The inability to obtain polyclonal enrichment using the conventional method could be due to the decrease of diversity during rescue resulting in the loss of the binders. The successful enrichment using the modified protocol is likely attributed to the better rescue diversity, thus making it better for monoclonal enrichment.

This highlights the difficulties associated to Fab panning using the conventional protocol. The coplasmid panning protocol was able to help improve the panning efficiency in this work. This is likely due to the presence of the pTUM4 coplasmid to aid not only in presentation but possibly in improved folding of the Fab. In summary, the coplasmid pTUM4 consisting of DsbA, DsbC, FkpA, and SurA is able to play an important role in improving the disulfide bond formation and Fab folding during phage packaging. This would provide an added advantage for Fab or other cysteine-based molecule panning experiments. Therefore, the efficiency of Fab presentation on phage particles can be improved with the assistance of this coplasmid system for phage display.

Acknowledgments

We thank Prof. Arne Skerra of Technische Universität München for providing the pTUM4 plasmid. The work was supported by the Malaysian Ministry of Education Fundamental Research Grant Scheme (Grant 203/CIPPM/6711473) and Malaysian Ministry of Education Higher Institution Centre of Excellence (HICoE) Grant (Grant 311/CIPPM/44001005).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ab.2015.02.026>.

References

- [1] H.R. Hoogenboom, A.D. Griffiths, K.S. Johnson, D.J. Chiswell, P. Hudson, G. Winter, Multi-subunit proteins on the surface of filamentous phage: methodologies for displaying antibody (Fab) heavy and light chains, *Nucleic Acids Res.* 19 (1991) 4133–4137.
- [2] C.F. Barbas, A.S. Kang, R.A. Lerner, S.J. Benkovic, Assembly of combinatorial antibody libraries on phage surfaces: the gene III site, *Proc. Natl. Acad. Sci. U.S.A.* 88 (1991) 7978–7982.
- [3] D. Missiakas, S. Raina, Protein folding in the bacterial periplasm, *J. Bacteriol.* 179 (1997) 2465.
- [4] M. Schlapschky, S. Grimm, A. Skerra, A system for concomitant overexpression of four periplasmic folding catalysts to improve secretory protein production in *Escherichia coli*, *Protein Eng. Des. Sel.* 19 (2006) 385–390.
- [5] V.E. Shevchik, G. Condemine, J. Robert-Baudouy, Characterization of DsbC, a periplasmic protein of *Erwinia chrysanthemi* and *Escherichia coli* with disulfide isomerase activity, *EMBO J.* 13 (1994) 2007.
- [6] M. Schlapschky, A. Skerra, Periplasmic chaperones used to enhance functional secretion of proteins in *E. coli*, *Methods Mol. Biol.* 705 (2011) 211–224.
- [7] J. Messens, J.F. Collet, K. Van Belle, E. Brosens, R. Loris, L. Wyns, The oxidase DsbA folds a protein with a nonconsecutive disulfide, *J. Biol. Chem.* 282 (2007) 31302–31307.
- [8] J.F. Collet, J.C. Bardwell, Oxidative protein folding in bacteria, *Mol. Microbiol.* 44 (2002) 1–8.
- [9] H. Nakamoto, J.C.A. Bardwell, Catalysis of disulfide bond formation and isomerization in the *Escherichia coli* periplasm, *Biochim. Biophys. Acta* 1694 (2004) 111–119.
- [10] J. Bardwell, J.-O. Lee, G. Jander, N. Martin, D. Belin, J. Beckwith, A pathway for disulfide bond formation in vivo, *Proc. Natl. Acad. Sci. U.S.A.* 90 (1993) 1038–1042.
- [11] A.R. Duguay, T.J. Silhavy, Quality control in the bacterial periplasm, *Biochim. Biophys. Acta* 1694 (2004) 121–134.
- [12] E. Bitto, D.B. McKay, Binding of phage-display-selected peptides to the periplasmic chaperone protein SurA mimics binding of unfolded outer membrane proteins, *FEBS Lett.* 568 (2004) 94–98.
- [13] G. Hennecke, J. Nolte, R. Volkmer-Engert, J. Schneider-Mergener, S. Behrens, The periplasmic chaperone SurA exploits two features characteristic of integral outer membrane proteins for selective substrate recognition, *J. Biol. Chem.* 280 (2005) 23540–23548.
- [14] J.G. Sklar, T. Wu, D. Kahne, T.J. Silhavy, Defining the roles of the periplasmic chaperones SurA, Skp, and DegP in *Escherichia coli*, *Genes Dev.* 21 (2007) 2473–2484.
- [15] W. Wang, S. Singh, D.L. Zeng, K. King, S. Nema, Antibody structure, instability, and formulation, *J. Pharm. Sci.* 96 (2007) 1–26.
- [16] Z. Sun, A. Almogren, P.B. Furtado, B. Chowdhury, M.A. Kerr, S.J. Perkins, Semi-extended solution structure of human myeloma immunoglobulin D determined by constrained X-ray scattering, *J. Mol. Biol.* 353 (2005) 155–173.
- [17] A. Sali, T.L. Blundell, Comparative protein modelling by satisfaction of spatial restraints, *J. Mol. Biol.* 234 (1993) 779–815.
- [18] J.D. Thompson, D.G. Higgins, T.J. Gibson, CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice, *Nucleic Acids Res.* 22 (1994) 4673–4680.

- [19] W. Humphrey, A. Dalke, K. Schulten, VMD: visual molecular dynamics, *J. Mol. Graph.* 14 (1996) 33–38.
- [20] D. Röthlisberger, A. Honegger, A. Plückthun, Domain interactions in the Fab fragment: a comparative evaluation of the single-chain Fv and Fab format engineered with variable domains of different stability, *J. Mol. Biol.* 347 (2005) 773–789.
- [21] M. Kirsch, M. Zaman, D. Meier, S. Dübel, M. Hust, Parameters affecting the display of antibodies on phage, *J. Immunol. Methods* 301 (2005) 173–185.
- [22] G. Soltes, M. Hust, K.K.Y. Ng, A. Bansal, J. Field, D.I.H. Stewart, S. Dübel, S. Cha, E.J. Wiersma, On the influence of vector design on antibody phage display, *J. Biotechnol.* 127 (2007) 626–637.
- [23] S.S. Sidhu, Engineering M13 for phage display, *Biomol. Eng.* 18 (2001) 57–63.
- [24] H. Bothmann, A. Plückthun, The periplasmic *Escherichia coli* peptidylprolyl cis, trans-isomerase FkpA. I. Increased functional expression of antibody fragments with and without cis-prolines, *J. Biol. Chem.* 275 (2000) 17100–17105.